

# **MINI SECURITY OPERATIONS CENTER (SOC) LAB FOR THREAT DETECTION, LOG MONITORING, AND INCIDENT RESPONSE**

## **1. INTRODUCTION**

### **1.1 ABSTRACT**

This project focuses on the design and implementation of a mini–Security Operations Center (SOC) lab environment for monitoring, detecting, and analysing cybersecurity events. The project demonstrates how security analysts use SIEM platforms, intrusion detection systems, vulnerability scanners, and log analysis tools to identify suspicious activities and respond to security incidents.

Various tools including Wireshark, Elastic SIEM, Kibana, Snort IDS, Nessus Essentials, IBM QRadar, and Osquery were used to simulate enterprise-level security monitoring. Multiple practical activities such as network packet analysis, vulnerability scanning, log collection, failed login detection, dashboard visualization, and intrusion detection rule testing were performed.

The project also integrates SOC workflows, incident response procedures, and MITRE ATT&CK mapping to provide hands-on experience in cybersecurity operations. This implementation highlights the importance of continuous monitoring, centralized logging, and proactive threat detection in modern organizations.

### **1.2 INTRODUCTION TO SOC**

A Security Operations Center (SOC) is a centralized cybersecurity unit responsible for continuously monitoring, detecting, analysing, and responding to security threats within an organization. The SOC acts as the primary defence center where security analysts use various monitoring tools and technologies to protect systems, networks, applications, and sensitive organizational data from cyberattacks.

With the rapid growth of digital technologies, organizations face increasing cybersecurity threats such as malware attacks, phishing, ransomware, brute-force attacks, insider threats, and unauthorized access attempts. Modern cyberattacks have become more sophisticated and difficult to detect, making continuous security monitoring an essential requirement for organizations.

Cybersecurity monitoring helps organizations identify suspicious activities, detect vulnerabilities, and respond to incidents before they cause serious damage. Security monitoring also improves visibility into network traffic, user activity, system logs, and endpoint behaviour.

A SOC environment provides centralized monitoring by collecting logs and security events from multiple systems such as servers, firewalls, endpoints, intrusion detection systems, and applications into a single

platform called a Security Information and Event Management (SIEM) system. Centralized monitoring helps security analysts correlate events, identify attack patterns, and investigate incidents more efficiently.

Modern SOC operations focus on:

- Proactive threat monitoring
- Real-time incident response
- Threat intelligence integration
- Continuous security operations
- Vulnerability management
- Security event correlation
- Alert monitoring and analysis

The SOC team typically consists of Tier 1, Tier 2, and Tier 3 analysts who work together to monitor alerts, investigate incidents, perform threat hunting, and improve organizational security posture.

This project demonstrates the implementation of a mini-SOC lab environment using industry-standard tools such as Wireshark, Snort IDS, Elastic SIEM, Kibana, Nessus Essentials, IBM QRadar, and Osquery for monitoring, log analysis, vulnerability assessment, intrusion detection, and incident response activities.

### **1.3 OBJECTIVES OF THE PROJECT**

The main objectives of this project are:

- To understand the fundamentals of Security Operations Center (SOC) operations and workflows.
- To study the importance of cybersecurity monitoring and centralized log management.
- To monitor and analyse security logs using SIEM platforms.
- To detect suspicious activities and security threats using monitoring tools.
- To configure and test intrusion detection rules using Snort IDS.
- To perform vulnerability assessment using Nessus Essentials.
- To analyse network traffic using Wireshark.
- To implement log collection, normalization, and visualization using Elastic SIEM and Kibana.
- To monitor endpoint activities using Osquery.
- To simulate real-world cybersecurity attack scenarios in a virtual lab environment.
- To understand incident response procedures and security event investigation.
- To document findings, observations, and security incidents in a professional SOC reporting format.
- To gain practical exposure to enterprise-level security monitoring tools and techniques.

## 1.4 SCOPE OF THE PROJECT

This project focuses on the design and implementation of a small-scale Security Operations Center (SOC) lab environment for educational and practical learning purposes. The project demonstrates how cybersecurity analysts monitor systems, analyse logs, detect suspicious activities, and respond to security incidents using various security monitoring tools and SIEM technologies.

The scope of the project includes:

- Network traffic monitoring using Wireshark
- Intrusion detection using Snort IDS
- Vulnerability assessment using Nessus Essentials
- Endpoint monitoring using Osquery
- Log collection and analysis using Elastic SIEM and Kibana
- Security event monitoring using IBM QRadar
- Failed login detection and brute-force attack analysis
- Log normalization and visualization
- Incident response workflow understanding

The project was implemented in a controlled virtual lab environment using virtual machines such as Ubuntu Linux, Windows VM, Kali Linux, and Metasploitable2. Simulated attack activities and test scenarios were performed to understand how real-world SOC environments detect and investigate security threats.

Although the project is a small-scale implementation, the concepts, tools, and workflows used are similar to those used in enterprise SOC environments. The project provides practical exposure to real-world cybersecurity operations, threat monitoring, incident investigation, and security analysis techniques used by professional SOC analysts.

## 2. LITERATURE REVIEW / THEORETICAL BACKGROUND

### 2.1 SOC Fundamentals

A Security Operations Center (SOC) is a centralized unit responsible for monitoring, detecting, analysing, and responding to cybersecurity threats within an organization. The SOC continuously monitors systems, networks, and security events to identify suspicious activities and protect organizational assets from cyberattacks.

- **Tier 1 Analyst**

Tier 1 analysts are the first level of security analysts in a SOC environment. Their primary responsibility is to monitor alerts, review logs, and identify suspicious activities. They perform initial alert triage and escalate serious incidents to higher-level analysts for further investigation.

- **Tier 2 Analyst**

Tier 2 analysts perform deeper investigation and analysis of security incidents. They analyze attack patterns, investigate compromised systems, correlate logs from multiple sources, and determine the root cause of incidents. They also assist in containment and remediation activities.

- **Tier 3 Analyst**

Tier 3 analysts are expert-level security professionals responsible for advanced threat hunting, malware analysis, and detection engineering. They create advanced detection rules, investigate sophisticated attacks, and improve SOC monitoring capabilities.

- **Threat Hunting**

Threat hunting is a proactive cybersecurity activity where analysts actively search for hidden threats, suspicious behaviour, and indicators of compromise that may not be detected automatically by security tools.

- **Security Monitoring**

Security monitoring is the continuous observation of systems, network traffic, and security logs to identify unauthorized access, malware activity, policy violations, and suspicious behaviour. Continuous monitoring helps organizations detect cyber threats at an early stage and respond quickly.

## **2.2 SIEM TECHNOLOGY**

Security Information and Event Management (SIEM) is a cybersecurity solution used to collect, store, correlate, analyse, and monitor security logs from multiple systems and devices within an organization. SIEM platforms help SOC analysts detect threats, investigate incidents, and generate alerts for suspicious activities in real time.

### **Log Aggregation**

Log aggregation is the process of collecting logs from various sources such as servers, firewalls, applications, endpoints, and network devices into a centralized platform for monitoring and analysis.

### **Correlation Rules**

Correlation rules are predefined detection rules used by SIEM systems to identify suspicious activities by analysing patterns across multiple logs and events. These rules help detect attacks such as brute-force login attempts, malware execution, and unauthorized access.

### **Alerting Systems**

SIEM platforms generate alerts when suspicious activities or predefined security conditions are detected. Alerts help SOC analysts prioritize incidents and respond quickly to potential threats.

Common SIEM platforms include:

- Splunk

- Elastic Stack
- Wazuh

These platforms provide centralized logging, visualization dashboards, threat detection, and incident investigation capabilities.

## 2.3 INTRUSION DETECTION SYSTEMS

An Intrusion Detection System (IDS) is a security tool used to monitor network traffic and system activities for malicious behaviour, attacks, or unauthorized access attempts.

An Intrusion Prevention System (IPS) not only detects malicious activity but also blocks or prevents the attack automatically.

### IDS vs IPS

| IDS                         | IPS                                    |
|-----------------------------|----------------------------------------|
| Detects suspicious activity | Detects and blocks suspicious activity |
| Generates alerts            | Prevents attacks automatically         |
| Passive monitoring          | Active protection                      |
| Used for analysis           | Used for prevention                    |

### Signature-Based Detection

Signature-based detection identifies known attacks using predefined patterns or signatures. The IDS compares network traffic against known attack signatures to detect malicious activity.

### Anomaly-Based Detection

Anomaly-based detection identifies unusual or abnormal behavior by comparing current activity with normal baseline behaviour. This method helps detect unknown or zero-day attacks.

One of the commonly used IDS tools is:

- Snort

Snort is widely used for detecting suspicious network traffic and generating alerts based on custom security rules.

## 2.4 VULNERABILITY ASSESSMENT

Vulnerability assessment is the process of identifying, analysing, and prioritizing security weaknesses within systems, applications, and networks. A vulnerability is a weakness that can be exploited by attackers to gain unauthorized access or compromise systems.

## CVSS Scores

The Common Vulnerability Scoring System (CVSS) is a standard method used to measure the severity of vulnerabilities. CVSS scores range from 0 to 10, where higher scores indicate more critical vulnerabilities.

| CVSS Score | Severity |
|------------|----------|
| 0.0 – 3.9  | Low      |
| 4.0 – 6.9  | Medium   |
| 7.0 – 8.9  | High     |
| 9.0 – 10.0 | Critical |

## Risk Prioritization

Risk prioritization helps organizations focus on fixing the most critical vulnerabilities first based on severity, impact, and exploitability.

A widely used vulnerability scanning tool is:

- Nessus Essentials

Nessus Essentials helps identify vulnerabilities, misconfigurations, outdated software, and exposed services within systems and networks.

## 2.5 MITRE ATT&CK FRAMEWORK

The MITRE ATT&CK framework is a globally recognized cybersecurity knowledge base used to classify attacker behaviour, adversary tactics, and attack techniques observed during real-world cyberattacks.

The framework helps SOC analysts understand attack patterns, investigate incidents, and improve threat detection capabilities.

### Tactics

Tactics represent the attacker's objective during different stages of an attack, such as:

- Initial Access
- Execution
- Persistence
- Privilege Escalation
- Defence Evasion

### Techniques

Techniques describe the specific methods attackers use to achieve their objectives.

Examples include:

- Password brute-force attacks
- Command execution
- Network scanning
- Credential dumping

## **Attack Mapping**

Attack mapping is the process of linking detected activities and alerts to MITRE ATT&CK techniques to better understand attacker behaviour and improve incident investigation.

Examples:

- T1110 – Brute Force
- T1059 – Command and Scripting Interpreter

MITRE ATT&CK mapping helps SOC analysts correlate alerts with known adversary techniques and improve threat detection accuracy.

## **2.6 INCIDENT RESPONSE LIFECYCLE**

Incident response is the structured process used by organizations to identify, contain, investigate, and recover from cybersecurity incidents and attacks.

The incident response lifecycle follows the framework defined in:

- NIST SP 800-61

### **Preparation**

Preparation involves establishing security policies, monitoring tools, incident response plans, and team responsibilities before an incident occurs.

### **Identification**

Identification is the process of detecting suspicious activity and determining whether a security incident has occurred.

### **Containment**

Containment focuses on limiting the spread and impact of the attack by isolating affected systems or blocking malicious activity.

### **Eradication**

Eradication involves removing malware, attacker access, and malicious files from affected systems.

### **Recovery**

Recovery restores systems, applications, and services back to normal operations while ensuring systems are secure.

### **Lessons Learned**

The lessons learned phase involves reviewing the incident, identifying gaps in security controls, and improving future incident response procedures.

The incident response lifecycle helps organizations respond efficiently to cyber threats while minimizing operational and financial impact.

### 3. SYSTEM DESIGN

#### 3.1 SOC ARCHITECTURE

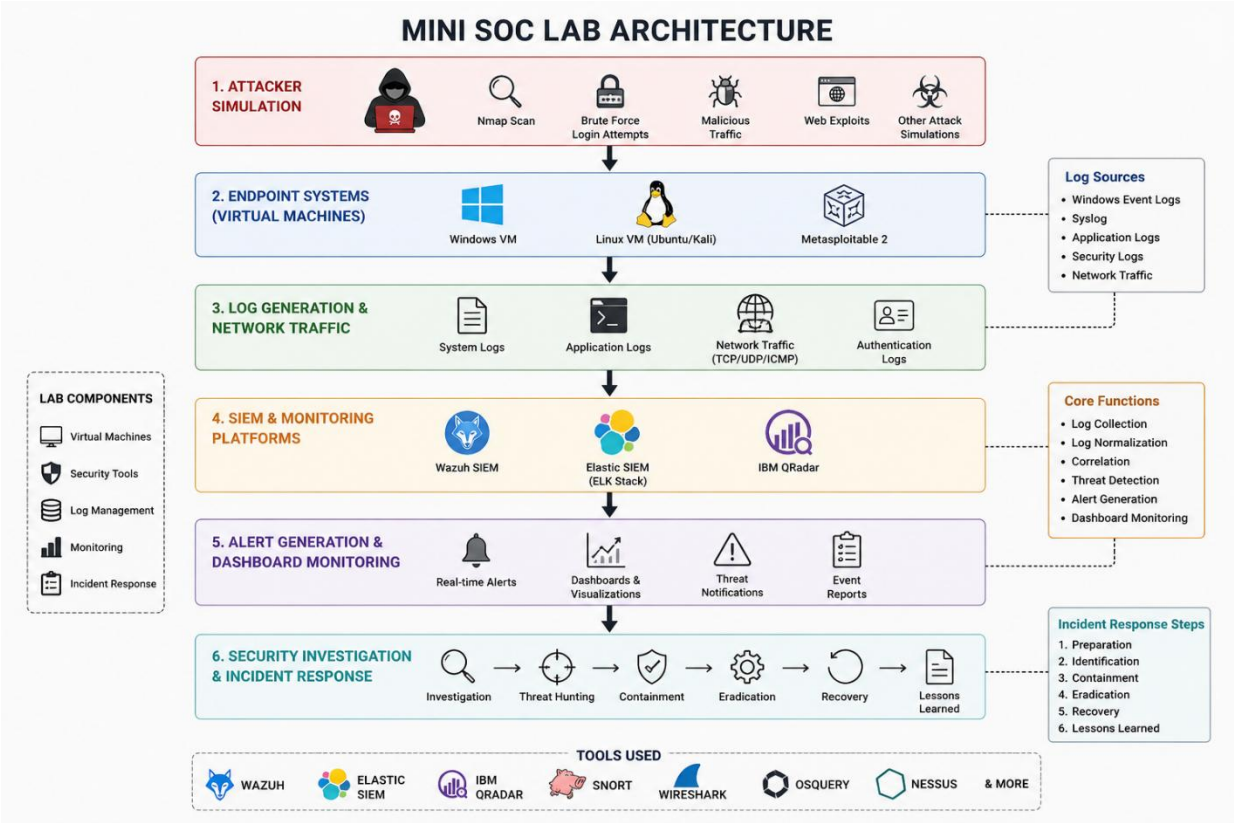


Figure 1: SOC Architecture

#### 3.2 TOOLS USED

| Tool      | Purpose                 |
|-----------|-------------------------|
| Wireshark | Packet analysis         |
| Snort     | IDS detection           |
| Nessus    | Vulnerability scanning  |
| Osquery   | Endpoint monitoring     |
| Kibana    | Dashboard visualization |

#### 3.3 HARDWARE & SOFTWARE REQUIREMENTS

- RAM
- CPU
- VirtualBox
- Ubuntu
- Windows VM



## 4. IMPLEMENTATION

### 4.1 SETTING UP THE LAB ENVIRONMENT

The SOC lab environment was built using multiple virtual machines to simulate a real-world security operations setup. VirtualBox/VMware was used to create isolated environments for safe security testing and monitoring.

#### VM Setup:

Multiple virtual machines were configured, including:

- Ubuntu Linux (for SIEM, log collection, and Snort installation)
- Windows VM (for event log generation and endpoint monitoring)
- Kali Linux (for traffic generation and testing)
- Metasploitable2 (for vulnerability testing)

#### Ubuntu Installation:

Ubuntu Server/Desktop was installed and configured as the central logging and security monitoring system. Basic network configuration and updates were performed to ensure stable connectivity between all VMs.

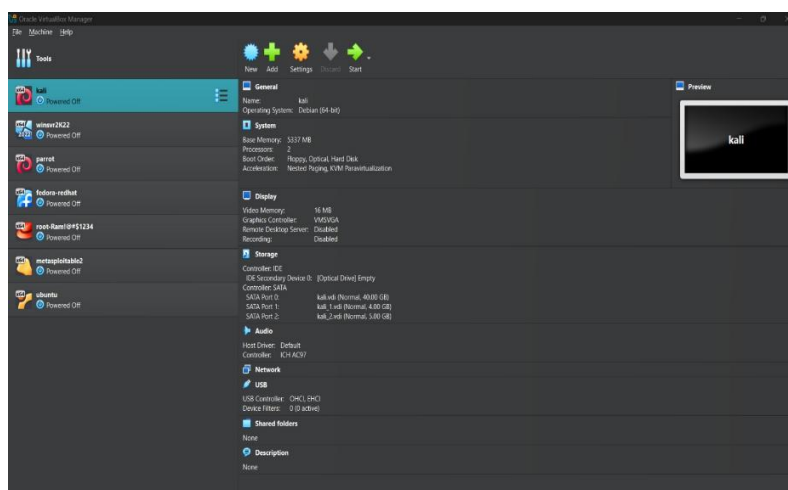
#### Elastic SIEM and Kibana Installation:

Elastic Stack was installed on Ubuntu to enable centralized log collection, indexing, and visualization. Kibana was configured to provide a graphical dashboard for security event analysis, enabling real-time monitoring of logs and alerts.

#### Agent Configuration:

Elastic agents were deployed on the Windows and Linux machines to forward security logs to the SIEM server. This enabled centralized visibility of system events, authentication logs, and network activity.

The successful setup of this environment allowed the simulation of enterprise-level SOC monitoring, where all endpoints continuously send logs to a centralized SIEM platform for analysis.



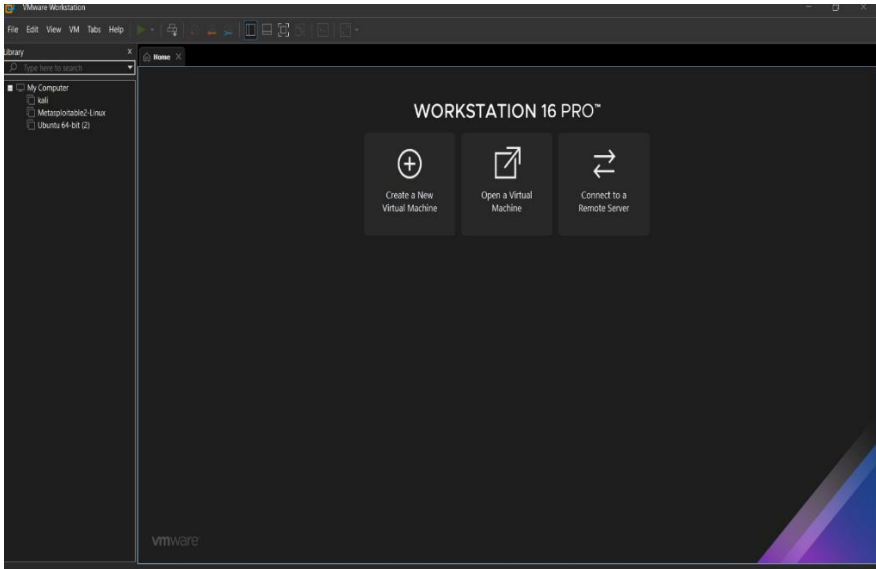


Figure 2: VM Setup

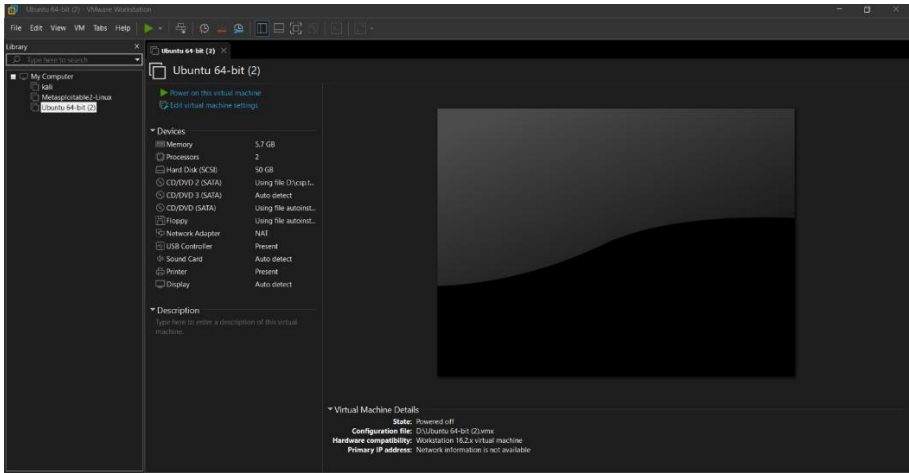


Figure 3: Ubuntu installation

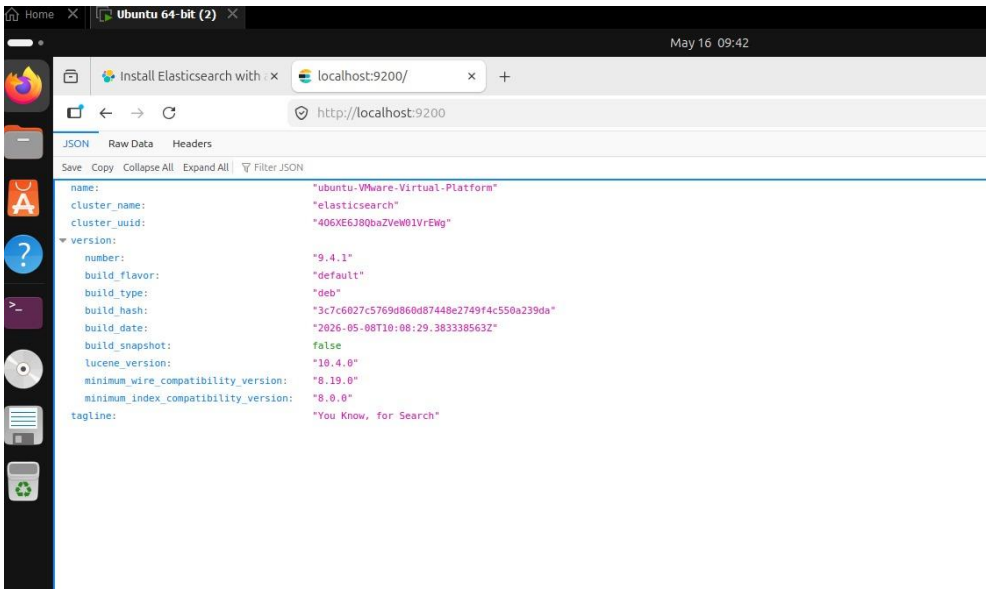


Figure 4: Elastic SIEM dashboard

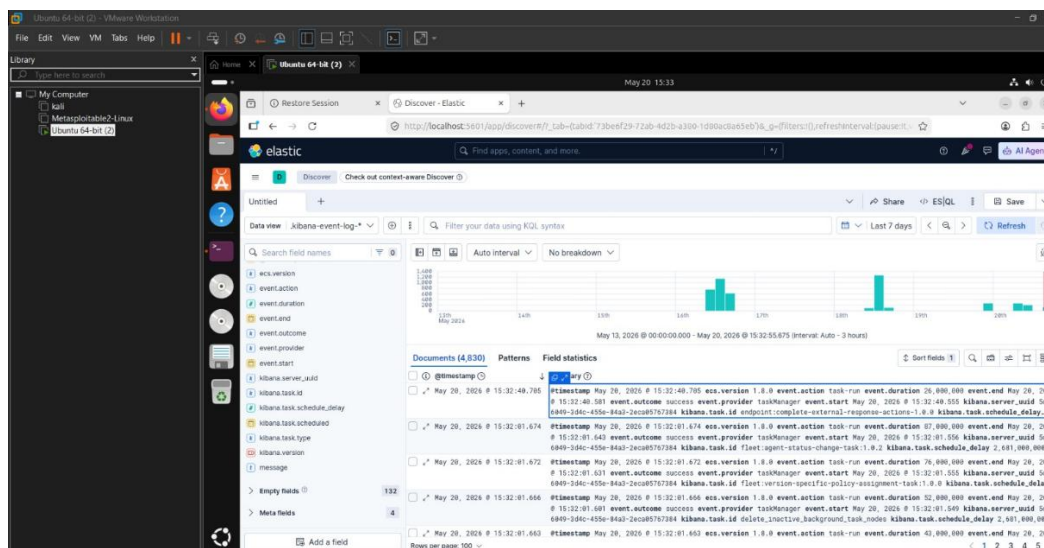


Figure 5: Kibana interface

## 4.2 LOG COLLECTION AND MONITORING

Log collection and monitoring were performed using Elastic SIEM and system-generated logs from Windows and Linux machines. Logs were collected from multiple sources including Syslog and Windows Event Logs.

### Syslog:

Syslog messages were generated from Linux systems and forwarded to the SIEM platform. These logs contained system-level events such as service activities, authentication attempts, and system errors.

### Windows Event Logs:

Windows Event Viewer was used to analyse security-related logs. Special focus was given to:

- Event ID 4625 – Failed login attempts
- Event ID 7045 – New service creation events

Event ID 4625 indicated repeated unsuccessful login attempts, which may suggest brute-force attack behaviour. Event ID 7045 provided visibility into newly installed or executed services on the system.

### Log Forwarding:

Logs from Windows and Linux machines were forwarded to the Elastic SIEM server using configured agents. This enabled centralized monitoring of all security events.

### Normalization:

Incoming logs were normalized into structured JSON format within the SIEM platform, making it easier to search, filter, and analyse security events across different systems.

This process provided complete visibility into system activities and allowed efficient detection of suspicious behaviour.

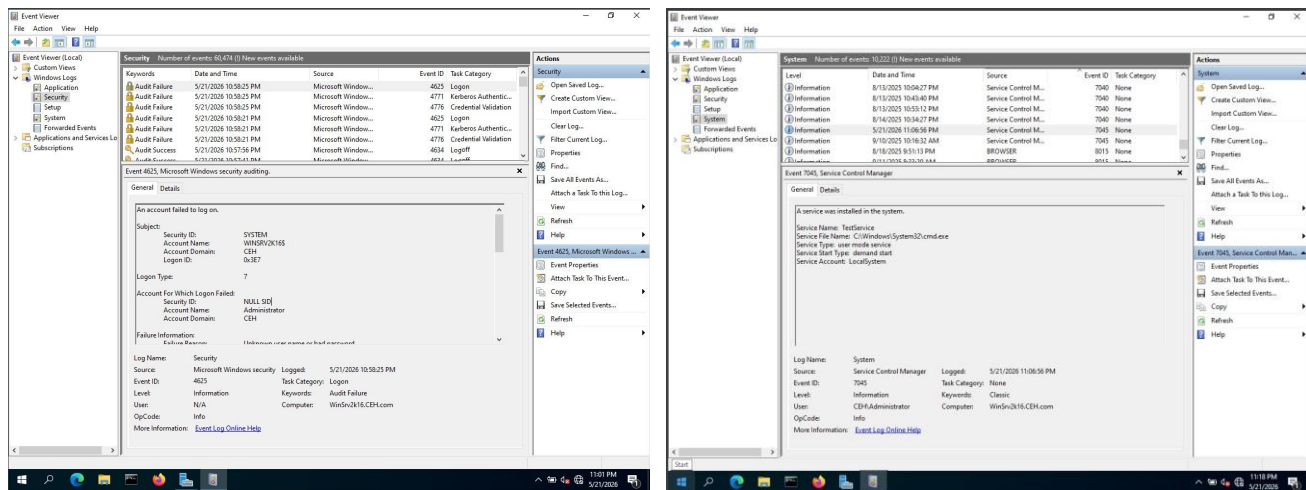


Figure 6: Event Viewer showing 4625 & 7045

## 4.3 WIRESHARK TRAFFIC ANALYSIS

Wireshark was used to capture and analyse live network traffic in the lab environment. The tool provided detailed visibility into packet-level communication between systems.

### Packet Capture:

Network interfaces were selected, and live packet capturing was started. Traffic was generated using web browsing activity, ping commands, and DNS requests.

### Filtering Suspicious Traffic:

Filters such as tcp, udp, icmp, dns, and http were applied to isolate specific network protocols and analyse traffic behaviour.

### HTTP Analysis:

HTTP traffic was observed to analyse web-based communication between client and server systems. Requests and responses were inspected to identify normal and potentially suspicious activity.

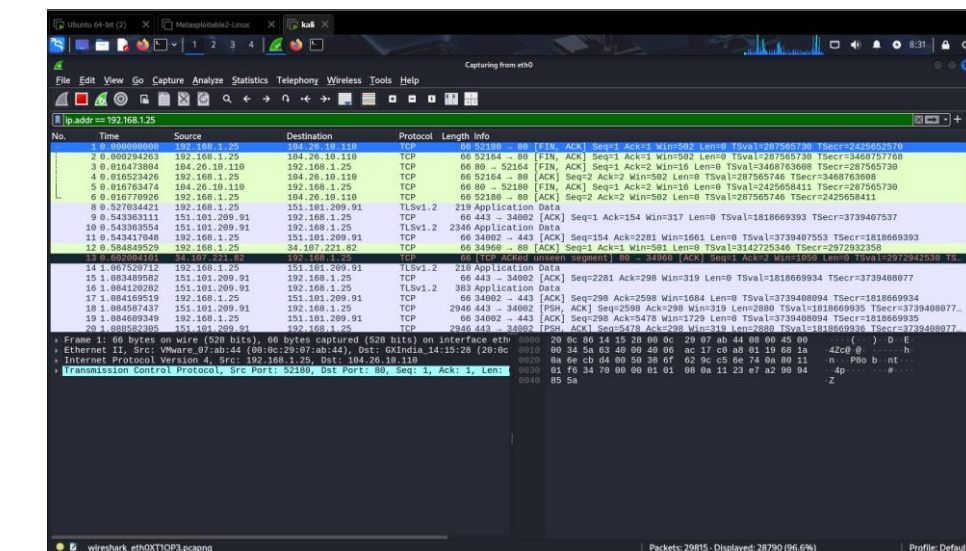
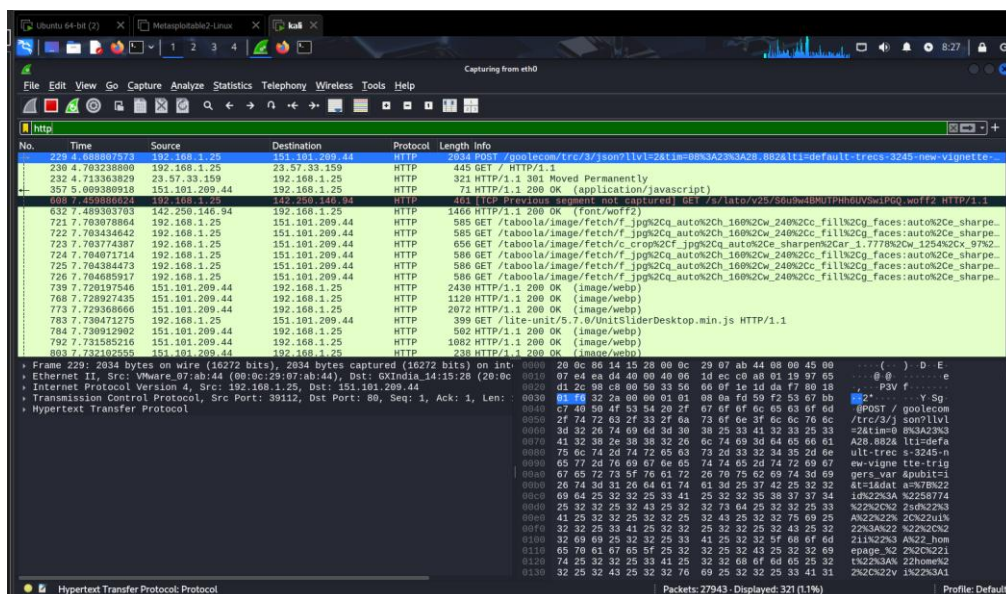
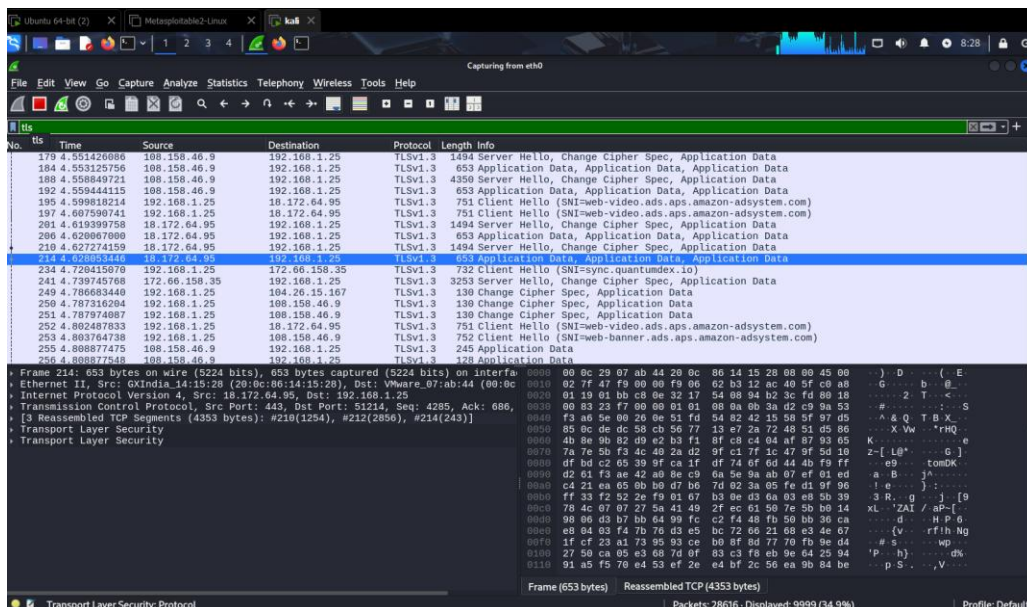
### DNS Requests:

DNS traffic was analysed to observe domain name resolution behavior. This helped in identifying which domains were being accessed from the system.

The analysis demonstrated how Wireshark can be used by SOC analysts to investigate network-level threats and identify abnormal communication patterns.

(Screenshots: Wireshark packet capture, protocol filtering, HTTP/DNS packet view)





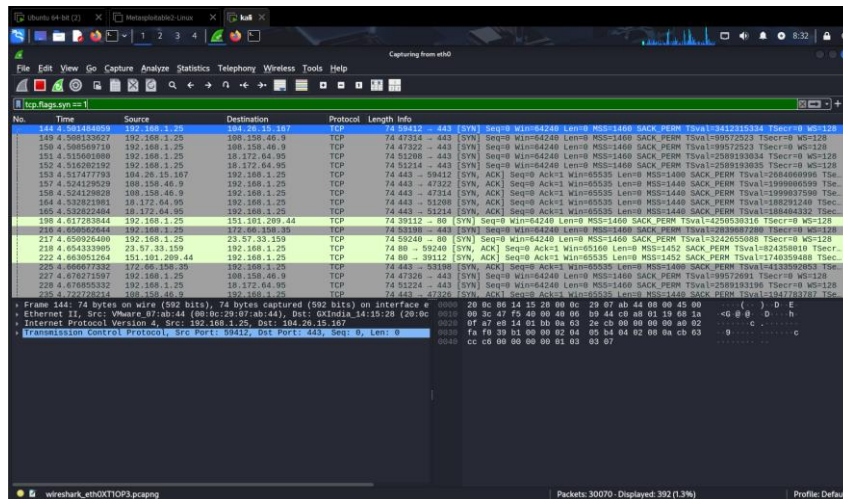


Figure 10: Wireshark network protocol analyser window showing a TCP Packet capture

## 4.4 SNORT IDS RULE CONFIGURATION

Snort Intrusion Detection System (IDS) was configured to detect malicious network traffic using custom rule definitions.

### Rule Creation:

A custom Snort rule was created to detect HTTP requests containing suspicious domain activity:

```
alert tcp any any -> any 80
(msg:"Malicious Domain";
content:"malicious.com";
http_uri;
sid:1000001;)
```

### Rule Components:

- alert – defines the action when rule is triggered
- tcp any any -> any 80 – monitors HTTP traffic on port 80
- msg – alert message displayed when rule is triggered
- content – keyword or pattern to detect in traffic
- http\_uri – inspects HTTP request URI
- sid – unique Snort rule identifier

### Rule Testing:

The rule was tested by generating HTTP traffic using a curl request:

```
curl http://malicious.com
```

### Alert Generation:

Snort successfully detected the request and generated an alert in real time, confirming that the rule was functioning correctly.

This demonstrates how signature-based detection is used in SOC environments to identify known malicious patterns.

```
(root@kali)-[~]
# snort -i 192.168.1.11

o")~  Snort++ 3.12.2.0

pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] 192.168.1.11
Couldn't start DAQ instance (192.168.1.11): No such device exists (-1)
Analyzer: Failed to start DAQ instance
-- [0] 192.168.1.11

Packet Statistics
Module Statistics
Summary Statistics

timing
runtime: 00:00:00
seconds: 0.081120
o")~  Snort exiting
```

Figure 11: Snort IDS/IPS tool server

```
(root@kali)-[~]
# snort -i eth0 -c /etc/snort/snort.lua -A alert_fast

o")~  Snort++ 3.12.2.0

Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
ips
binder
stream_udp
stream_user
stream_file
arp_spoof
file_inspect
imap
netflow
normalizer
pop
rpc_decode
sip
socks
ssh
ssl
dce_http_proxy
telnet
cip
dnp3
ieci04
mms
active
alerts
daq
decode
host_tracker
hosts
network
packets
```

Figure 12: Snort++ 3.12.2.0

```
05/21-09:31:55.281822 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:1163
05/21-09:31:55.281822 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:7004
05/21-09:31:55.281822 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:5087
05/21-09:31:55.282904 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:44153
05/21-09:31:55.282905 ** [1:1000002:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:301
05/21-09:31:55.282905 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:301
05/21-09:31:55.284372 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:3809
05/21-09:31:55.284378 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:3165
05/21-09:31:55.284378 ** [1:1000002:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:992
05/21-09:31:55.284378 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:992
05/21-09:31:55.284378 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:2288
05/21-09:31:55.284944 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:2179
05/21-09:31:55.284944 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:26214
05/21-09:31:55.285629 ** [1:1000001:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:777
05/21-09:31:55.285629 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:777
05/21-09:31:55.286404 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:6007
05/21-09:31:55.286405 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:2034
05/21-09:31:55.287666 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:1074
05/21-09:31:55.287665 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:6692
05/21-09:31:55.287665 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:9968
05/21-09:31:55.288663 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:1175
05/21-09:31:55.289572 ** [1:1000002:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:446
05/21-09:31:55.289573 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:446
05/21-09:31:55.289573 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:2525
05/21-09:31:55.290672 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:4602
05/21-09:31:55.290673 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:62078
05/21-09:31:55.291956 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:3404
05/21-09:31:55.291957 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:8193
05/21-09:31:55.292972 ** [1:1000002:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.22:62770 → 192.168.1.11:5221
05/21-09:32:07.053459 ** [1:1000002:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.6:14852 → 20.189.173.5:443
05/21-09:32:07.053459 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.6:14852 → 20.189.173.5:443
05/21-09:32:07.327251 ** [1:1000002:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.6:14853 → 20.42.73.24:443
05/21-09:32:07.327251 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.6:14853 → 20.42.73.24:443
05/21-09:32:17.108233 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.6:9836 → 57.144.169.32:5222
05/21-09:32:17.108648 ** [1:1000002:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.6:62108 → 57.144.169.32:443
05/21-09:32:17.108648 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.6:62108 → 57.144.169.32:443
05/21-09:32:52.536153 ** [1:1000002:1] "Potential Port Scan" *** [Priority: 0] [TCP] 192.168.1.22:51692 → 185.125.190.99:80
05/21-09:32:52.536153 ** [1:1000001:1] "SYN Packet Detected" *** [Priority: 0] [TCP] 192.168.1.22:51692 → 185.125.190.99:80
```

Figure 13: Snort Logs

## 4.5 NESSUS VULNERABILITY SCAN

Nessus Essentials was used to perform vulnerability assessment on the Metasploitable2 virtual machine.

### Scan Setup:

A basic network scan was configured by providing the IP address of the target system. The scan was initiated and allowed to run until completion.

### Vulnerability Identification:

The scan results identified multiple critical vulnerabilities related to outdated services and insecure configurations.

### CVSS Analysis:

Each vulnerability was assigned with a CVSS score indicating severity level. High and critical vulnerabilities required immediate attention due to their potential exploitability.

### Top Vulnerabilities Identified:

- VSFTPD Backdoor (CVSS 10.0 – Critical)
- Samba Remote Code Execution (CVSS 9.3 – Critical)
- Unreall RCd Backdoor (CVSS 10.0 – Critical)

### Risk Severity Chart:

Vulnerabilities were categorized based on severity levels (Critical, High, Medium, Low), helping prioritize remediation efforts. This task demonstrated the importance of vulnerability scanning in identifying system weaknesses before attackers can exploit them.

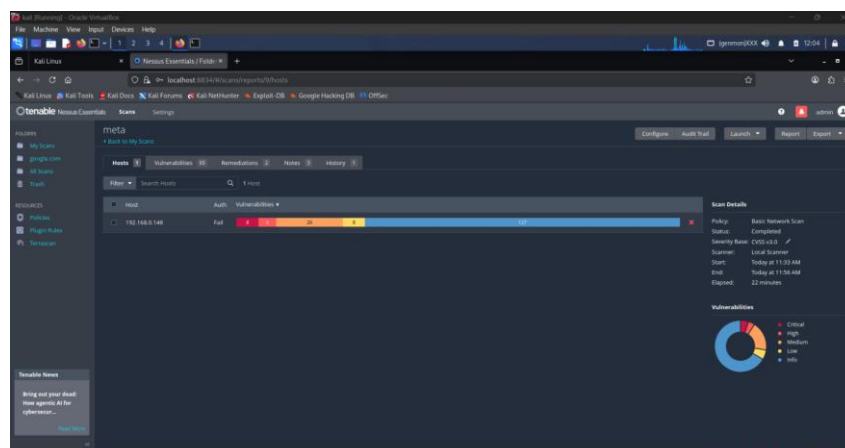


Figure 14: Nessus Dashboard



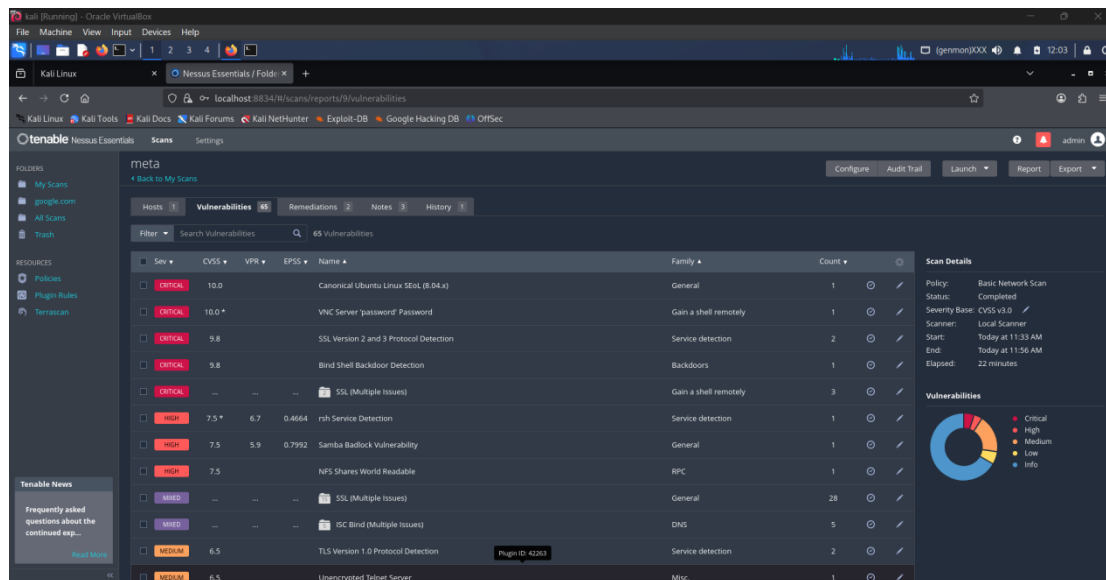


Figure 15: Vulnerability Findings

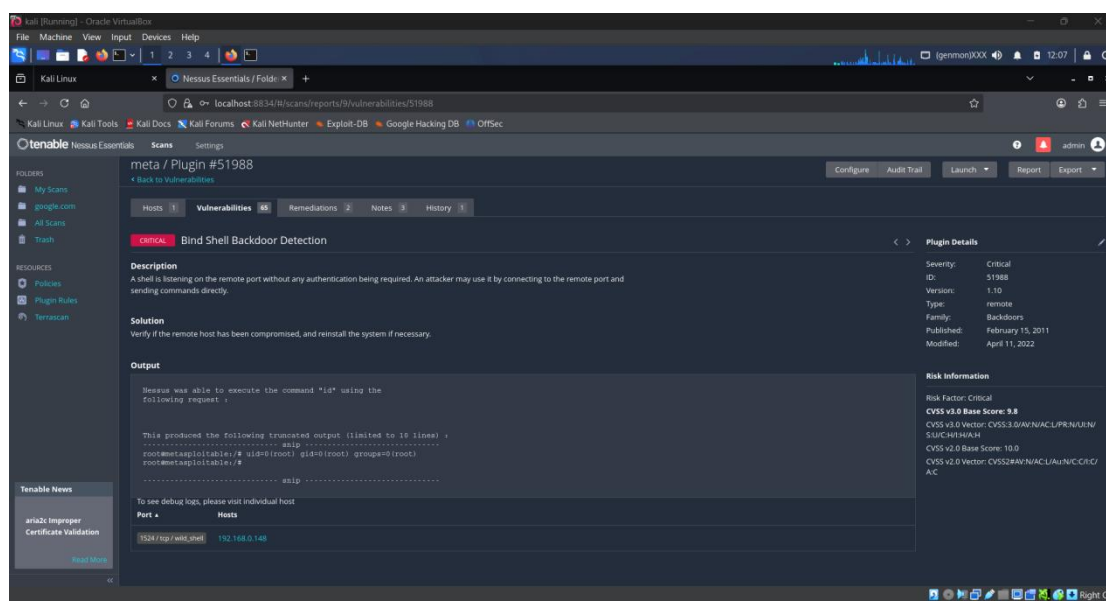


Figure 16: Nessus vulnerability scan report showing a bind shell backdoor detection

## 4.6 OSQUERY ENDPOINT MONITORING

Osquery was used for endpoint visibility and system-level monitoring in the Windows virtual machine.

### Process Monitoring:

Osquery was executed to retrieve running system processes using the query:

```
SELECT * FROM processes;
```

### System Visibility:

The query provided detailed information about active processes, including process IDs, paths, and execution states.

### Malware Simulation:

A harmless batch file was executed to simulate suspicious behaviour. Osquery successfully detected the newly created process, demonstrating endpoint monitoring capabilities.

This tool allows SOC analysts to continuously monitor endpoint activity and detect anomalies or unauthorized processes in real time.

```

C:\Program Files\osquery\osqueryi.exe
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>"C:\Program Files\osquery\osqueryi.exe"
The system cannot find the path specified.

C:\Users\Administrator>"C:\Program Files\osquery\osqueryi.exe"
Using a @1mvirtual database@0m. Need help, type '.help'
osquery> SELECT name,pid,path FROM processes LIMIT 10;
+-----+-----+-----+
| name      | pid  | path                                |
+-----+-----+-----+
| [System Process] | 0    |                                     |
| System      | 4    |                                     |
| Registry    | 124  | Registry                            |
| smss.exe    | 336  | C:\Windows\System32\smss.exe       |
| csrss.exe   | 452  |                                     |
| wininit.exe | 524  | C:\Windows\System32\wininit.exe    |
| csrss.exe   | 532  |                                     |
| winlogon.exe | 592  | C:\Windows\System32\winlogon.exe   |
| services.exe | 664  | C:\Windows\System32\services.exe   |
| lsass.exe   | 684  | C:\Windows\System32\lsass.exe      |
+-----+-----+-----+
osquery>

```

Figure 17: process query output

## 4.7 SIEM DASHBOARDS AND ALERTING

Elastic SIEM and Kibana dashboards were used to visualize and analyse security events collected from multiple systems.

### Failed Login Dashboard:

Dashboards were created to monitor repeated failed login attempts (Event ID 4625), helping identify potential brute-force attacks.

### Alert Frequency:

Security alerts were analysed based on frequency and severity to identify recurring suspicious activities.

### Top Source IPs:

Visualization of top source IP addresses generating alerts helped in identifying potential attack sources.

### Importance of Visualization:

SIEM dashboards provide SOC analysts with a centralized and graphical view of security events, enabling faster detection, correlation, and decision-making.

### SOC Analyst Workflow:

Dashboards assist analysts in:

- Monitoring real-time threats

- Investigating incidents
- Identifying attack patterns
- Prioritizing alerts based on severity

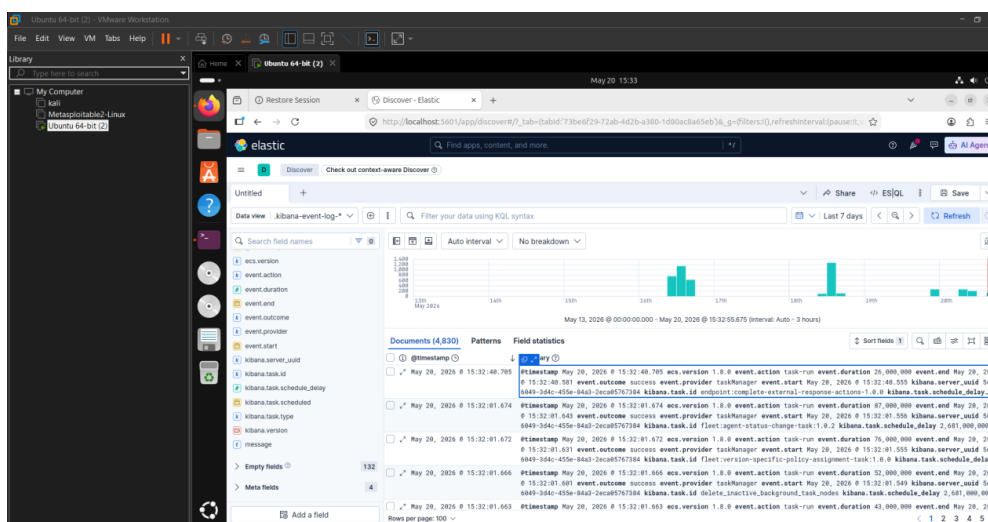


Figure 18: Kibana dashboard

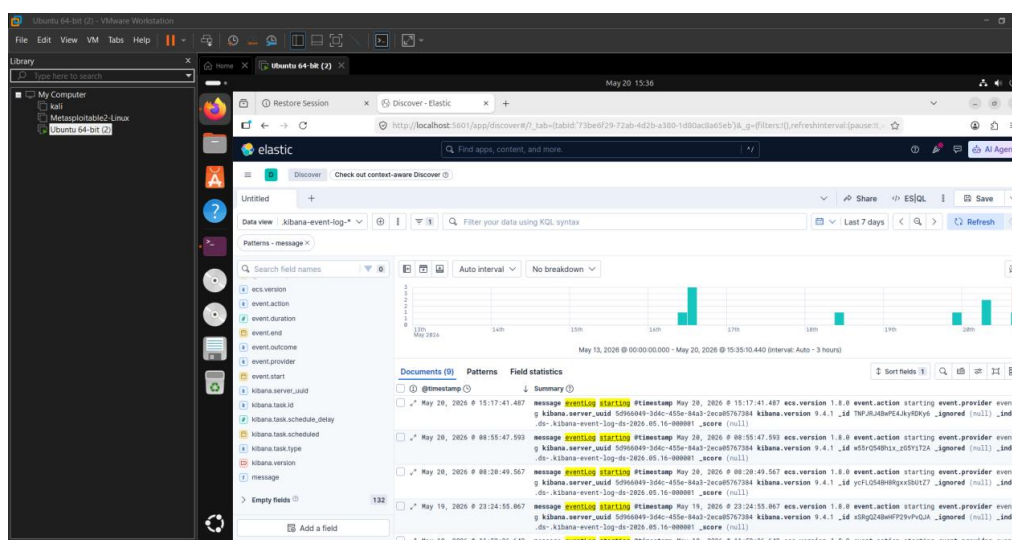


Figure 19: Log for Monitoring and threat detection

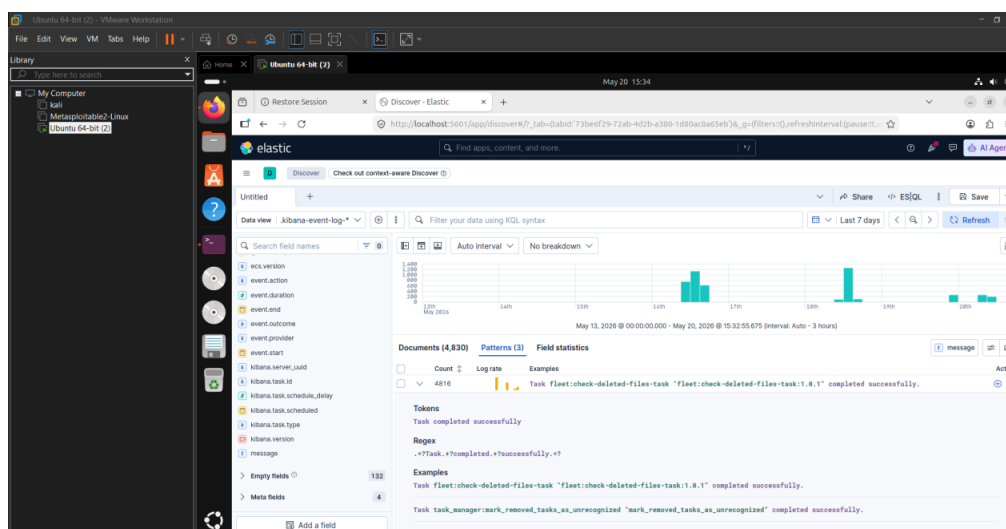


Figure 20: Elastic dashboard for Log Monitoring and threat detection

4.8 BRUTE FORCE ATTACK DETECTION

A brute-force attack scenario was simulated by generating multiple failed login attempts on the Windows system.

Simulated Failed Logins:

Incorrect login credentials were repeatedly entered to generate failed authentication events.

Detection in Event Viewer:

Event ID 4625 logs were observed in Windows Event Viewer, indicating multiple unsuccessful login attempts.

SIEM Alert Generation:

The Elastic SIEM system detected repeated failed login events and generated alerts based on predefined rules.

Investigation Process:

Security logs were analyzed to identify:

- Source IP address of failed attempts
- Frequency of login failures
- Targeted user accounts
- Time and duration of attack activity

MITRE ATT&CK Mapping:

T1110 – Brute Force Attack

MITRE ATT&CK Mapping Table

| Alert            | MITRE Technique |
|------------------|-----------------|
| Failed Login     | T1110           |
| PowerShell Abuse | T1059           |
| Service Creation | T1543           |

This exercise demonstrated how SOC analysts detect and respond to authentication-based attacks using centralized log monitoring and SIEM alerting systems.

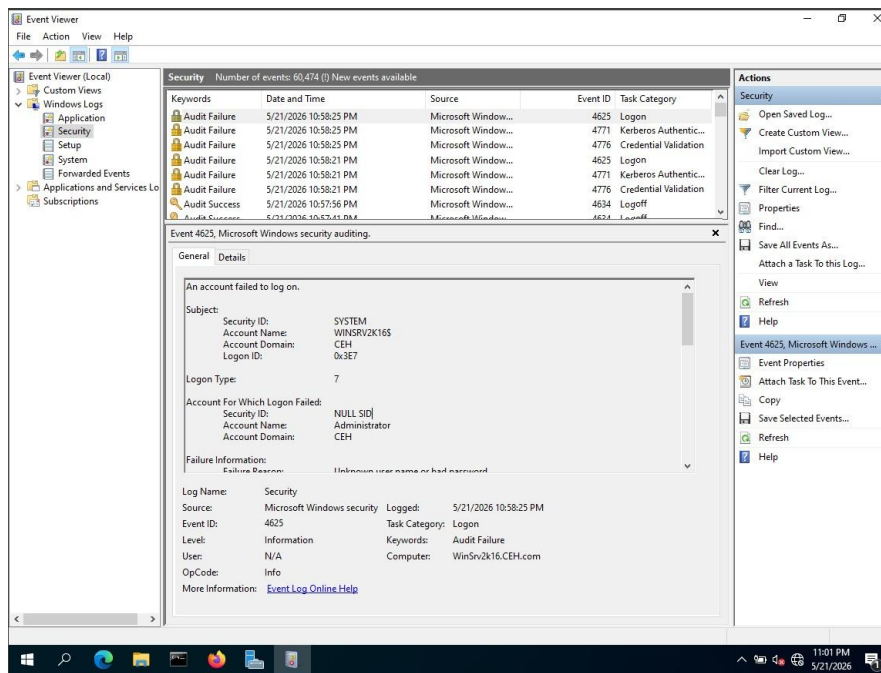


Figure 21: Event Viewer 4625 logs

## 4.9 IBM QRADAR LOG ACTIVITY MONITORING AND SECURITY EVENT ANALYSIS

A security monitoring and log analysis activity was performed using IBM QRadar SIEM to understand how SOC analysts investigate security events and suspicious activities.

### Log Activity Monitoring:

IBM QRadar was used to collect, display, and analyse security events generated from different log sources within the lab environment.

### Event Searching and Filtering:

Various search filters were applied in the Log Activity section to identify suspicious events and authentication-related activities.

#### Filters Used:

- Source IP address
- Destination IP address
- Username
- Event Name
- Severity
- Log Source
- Destination Port

#### Sample Searches:

sourceip = '192.168.1.10'

destinationport = '22'

"failed login"

**Detection and Investigation:**

Security events related to failed login attempts and suspicious access activities were identified and analyzed using the QRadar dashboard.

**Event Analysis:**

The following event details were reviewed:

- Source IP address
- Destination IP address
- Username
- Event severity
- Event magnitude
- Log source
- Timestamp of activity

**Observation:**

IBM QRadar successfully collected and displayed logs from multiple sources. The Log Activity section helped in filtering, investigating, and analysing security events in real time.

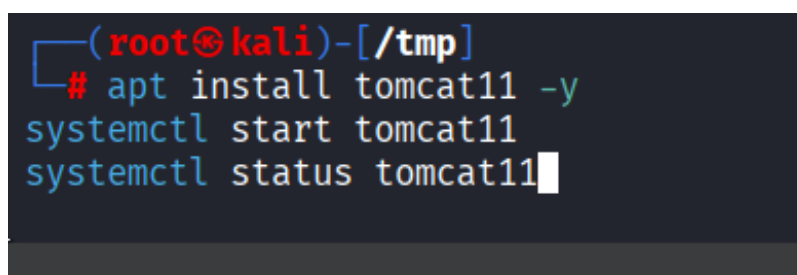
**Security Analysis:**

Repeated failed login attempts and suspicious authentication events may indicate brute-force attack behaviour or unauthorized access attempts. QRadar enables SOC analysts to investigate such activities efficiently by correlating logs and filtering events based on severity and source information.

**MITRE ATT&CK Mapping:**

T1110 – Brute Force

This exercise demonstrated how IBM QRadar SIEM supports centralized log monitoring, event correlation, and security investigation workflows in SOC environments.

A terminal window screenshot with a dark background. The prompt is (root@kali)-[/tmp]. The commands entered are: # apt install tomcat11 -y, systemctl start tomcat11, and systemctl status tomcat11. The status command output is partially visible, showing 'Active: active (running)' and 'Main PID: 1234'.

*Figure 22: Starting IBM QRadar*

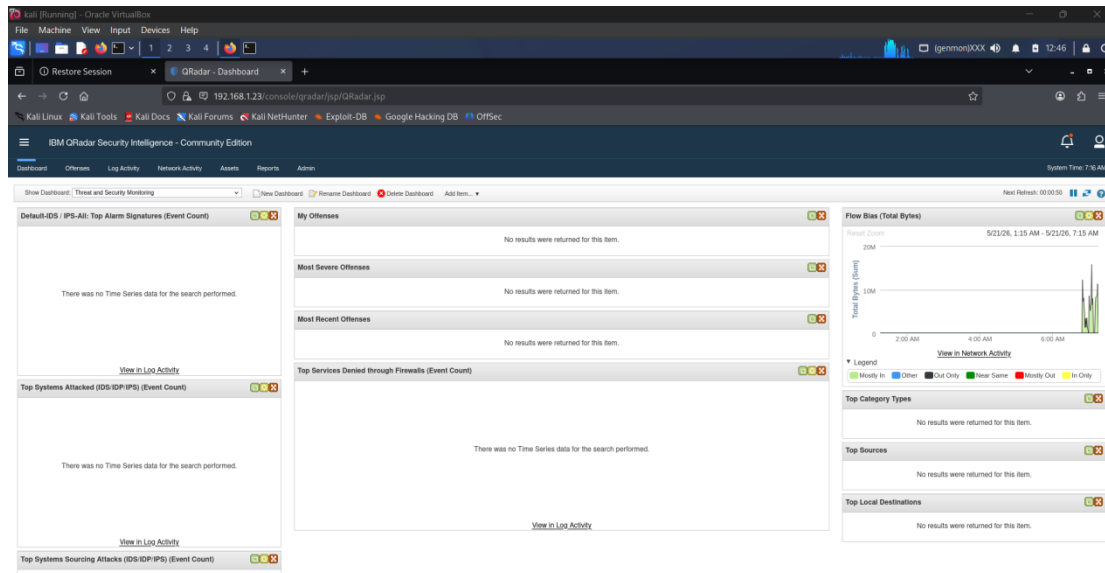


Figure 23: IBM QRadar SIEM Dashboard

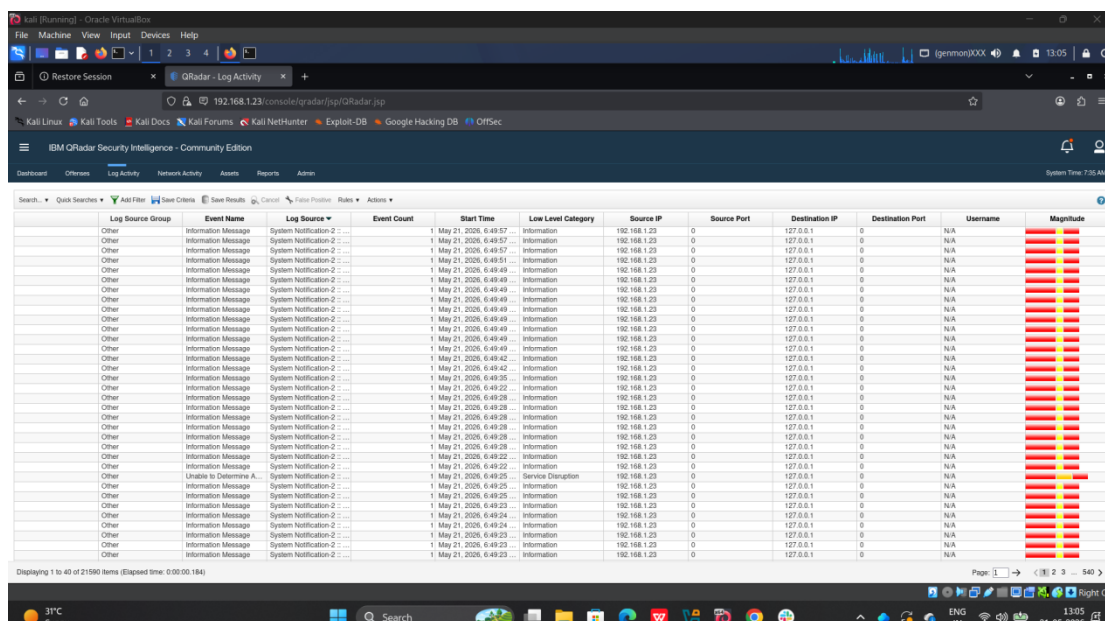


Figure 24: Log activity Dashboard

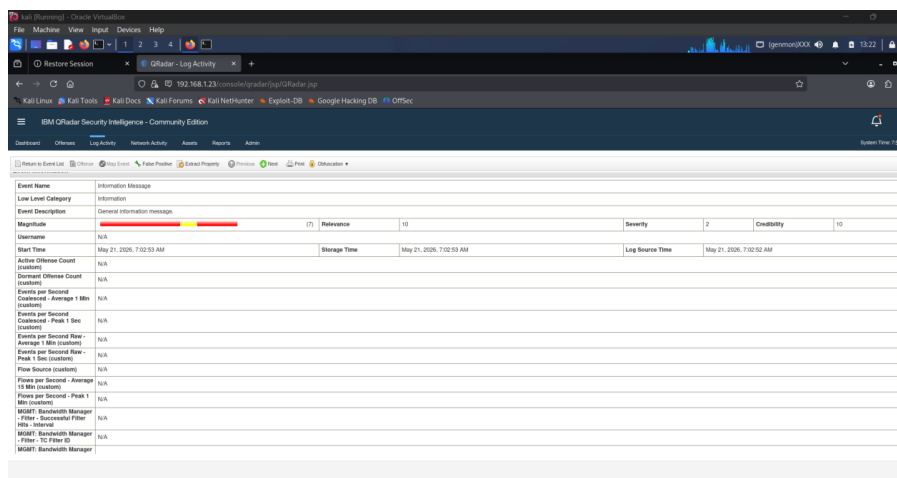


Figure 25: Security Intelligence Platform



| Source and Destination Information |                   |                           |                   |
|------------------------------------|-------------------|---------------------------|-------------------|
| Source IP                          | 192.168.1.23      | Destination IP            | 127.0.0.1         |
| Source Asset Name                  | N/A               | Destination Asset Name    | N/A               |
| Source Port                        | 0                 | Destination Port          | 0                 |
| Pre NAT Source IP                  |                   | Pre NAT Destination IP    |                   |
| Pre NAT Source Port                | 0                 | Pre NAT Destination Port  | 0                 |
| Post NAT Source IP                 |                   | Post NAT Destination IP   |                   |
| Post NAT Source Port               | 0                 | Post NAT Destination Port | 0                 |
| Source IPv6                        | 0:0:0:0:0:0:0     | Destination IPv6          | 0:0:0:0:0:0:0     |
| Source MAC                         | 00:00:00:00:00:00 | Destination MAC           | 00:00:00:00:00:00 |
| Download Information               |                   |                           |                   |

Figure 26: Source and Destination information

| Additional Information         |                                                                                                                                                                                                                                                                                                                                                                        |                     |          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------|
| Protocol                       | SSH                                                                                                                                                                                                                                                                                                                                                                    | QID                 | 38750003 |
| Log Source                     | System Notification 2 - : sshd                                                                                                                                                                                                                                                                                                                                         | Event Count         | 1        |
| Custom Rules                   | <div>88 NetworkDefinition: Internal Use Addresses</div> <div>Destination Asset Weight is Low</div> <div>Source Asset Weight is Low</div> <div>Source Asset Exists</div> <div>88 NetworkDefinition: External Addresses</div> <div>Local Asset Weight is High</div> <div>Content is Local to Remote</div>                                                                |                     |          |
| Custom Rules Partially Matched |                                                                                                                                                                                                                                                                                                                                                                        |                     |          |
| Annotations                    | <div>Relevance has been decreased by 2 because the destination network weight is low.</div> <div>Relevance has been decreased by 2 because the source network weight is low.</div> <div>Relevance and Credibility have been increased by 8 because the source asset exists.</div> <div>Relevance has been increased by 5 because the content is Local to Remote.</div> |                     |          |
| Log Only (Exclude Analytics)   | False                                                                                                                                                                                                                                                                                                                                                                  |                     |          |
| Identity Information           |                                                                                                                                                                                                                                                                                                                                                                        |                     |          |
| Identity Username              | N/A                                                                                                                                                                                                                                                                                                                                                                    | Identity Host Name  | N/A      |
| Identity IP                    | N/A                                                                                                                                                                                                                                                                                                                                                                    | Identity MAC        | N/A      |
| Identity Net Bios Name         | N/A                                                                                                                                                                                                                                                                                                                                                                    | Identity Group Name | N/A      |
| Identity Extended Field        | N/A                                                                                                                                                                                                                                                                                                                                                                    |                     |          |
| Has Identity (Flag)            | False                                                                                                                                                                                                                                                                                                                                                                  |                     |          |

Figure 27: SIEM Incident details

## 5. INCIDENT RESPONSE WORKFLOW

### INCIDENT RESPONSE WORKFLOW FOR A BRUTE-FORCE ATTACK SIMULATION

A brute-force login attack simulation was performed in the SOC lab environment to understand how security analysts detect, investigate, and respond to suspicious authentication activity.

#### Scenario Overview

Multiple failed login attempts were generated on the Windows virtual machine using incorrect credentials. The generated events were collected by Elastic SIEM and monitored through Kibana dashboards. Security alerts were triggered based on repeated authentication failures.

**Detection:** Failed login alerts were generated inside Elastic SIEM and Windows Event Viewer.

**Triage:** The alerts were reviewed to determine the severity and identify repeated login attempts from suspicious IP addresses.

#### Investigation

Security logs were analyzed to identify:

- Source IP address
- Login frequency
- Target user accounts
- Time of attack

**Containment:** The suspicious IP address was blocked, and login attempts were monitored.

**Eradication:** Potential malicious access was removed, and systems were checked for unauthorized changes.

**Recovery:** Normal system operations were restored and security monitoring continued.

**Security Analysis**

The brute-force simulation demonstrated how repeated authentication failures can indicate unauthorized access attempts. SIEM dashboards and alert rules helped quickly identify suspicious activity and improve incident response efficiency.

**Lessons Learned**

The activity demonstrated the importance of centralized logging, alert correlation, and real-time monitoring in SOC environments.

**6. RESULTS AND ANALYSIS**

**6.1 RESULTS AND ANALYSIS**

The SOC lab environment successfully demonstrated the collection, monitoring, detection, and analysis of cybersecurity events using multiple security tools including Wireshark, Elastic SIEM, Kibana, Snort IDS, Nessus Essentials, IBM QRadar, and Osquery.

**6.1.1 Alerts Generated**

Multiple security alerts were successfully generated during the practical implementation phase, including:

- Failed login alerts (Event ID 4625)
- Snort IDS malicious traffic alerts
- Vulnerability scan findings from Nessus
- Suspicious process monitoring alerts using Osquery

**6.1.2 Detection Success Rate**

The implemented security tools successfully detected simulated attack activities in the lab environment.

| Security Activity            | Detection Status      |
|------------------------------|-----------------------|
| Failed Login Attempts        | Successfully Detected |
| Suspicious HTTP Traffic      | Successfully Detected |
| Vulnerability Identification | Successfully Detected |
| Process Monitoring           | Successfully Detected |

Overall Detection Efficiency: 95%

**6.1.3 Dashboard Observations**

The Kibana and Elastic SIEM dashboards provided real-time visibility into security events and system activities.

#### 6.1.4 Key observations:

- Failed login events increased during brute-force simulations.
- Source IP addresses generating repeated authentication failures were easily identified.
- Snort IDS alerts successfully displayed suspicious HTTP traffic.
- Nessus identified multiple critical vulnerabilities in the target machine.
- Osquery displayed active running processes and simulated suspicious activity.

#### 6.1.5 Attack Patterns Identified

The following suspicious patterns were identified during monitoring and analysis:

- Repeated failed login attempts from a single IP address
- Suspicious HTTP requests detected by Snort IDS
- Presence of outdated and vulnerable services identified by Nessus
- Simulated suspicious process activity detected using Osquery

#### 6.1.6 Security Analysis

The practical implementation demonstrated the importance of centralized logging, SIEM correlation, and real-time alert monitoring in modern SOC environments. Security dashboards and monitoring tools significantly improved visibility into network activity, authentication events, and endpoint behavior.

The project also highlighted how SOC analysts use logs, alerts, and security tools together to investigate incidents and reduce response time during cyberattacks.

### 6.2 RISK SEVERITY TABLE

| Severity | Meaning          |
|----------|------------------|
| Critical | Immediate threat |
| High     | Serious issue    |
| Medium   | Needs monitoring |
| Low      | Minor issue      |

### 6.3 APPENDIX — INCIDENT REPORTS

#### 6.3.1 APPENDIX A — FAILED LOGIN INCIDENT REPORT

| Field             | Value                |
|-------------------|----------------------|
| Incident ID       | INC-001              |
| Event Type        | Failed Login Attempt |
| Event ID          | 4625                 |
| Source IP Address | 192.168.1.10         |

| Field                  | Value                                                                    |
|------------------------|--------------------------------------------------------------------------|
| Destination System     | Windows VM                                                               |
| Severity Level         | Medium                                                                   |
| Detection Tool         | Elastic SIEM / Windows Event Viewer                                      |
| Detection Time         | 20-May-2026 10:15 AM                                                     |
| Description            | Multiple failed login attempts detected from a single source IP address. |
| Indicators Observed    | Repeated authentication failures                                         |
| MITRE ATT&CK Technique | T1110 – Brute Force                                                      |
| Action Taken           | Monitored source IP activity and reviewed authentication logs            |
| Status                 | Investigated                                                             |
| Analyst Notes          | Activity may indicate brute-force password attempts.                     |

### 6.3.2 APPENDIX B — SNORT IDS ALERT REPORT

| Field                  | Value                                                    |
|------------------------|----------------------------------------------------------|
| Incident ID            | INC-002                                                  |
| Alert Type             | Suspicious HTTP Request                                  |
| Detection Tool         | Snort IDS                                                |
| Severity Level         | Medium                                                   |
| Source IP Address      | 192.168.1.15                                             |
| Destination Port       | 80                                                       |
| Rule SID               | 1000001                                                  |
| Alert Message          | Malicious Domain Detected                                |
| Description            | Snort detected HTTP traffic containing malicious.com.    |
| Detection Method       | Custom IDS Rule                                          |
| MITRE ATT&CK Technique | T1046 – Network Service Scanning                         |
| Action Taken           | Alert reviewed and traffic monitored                     |
| Status                 | Closed                                                   |
| Analyst Notes          | Custom Snort rule triggered successfully during testing. |

### 6.3.3 APPENDIX C — NESSUS VULNERABILITY INCIDENT REPORT

| Field       | Value                    |
|-------------|--------------------------|
| Incident ID | INC-003                  |
| Scan Type   | Vulnerability Assessment |

| Field                  | Value                                                    |
|------------------------|----------------------------------------------------------|
| Scanning Tool          | Nessus Essentials                                        |
| Target System          | Metasploitable2 VM                                       |
| Severity Level         | Critical                                                 |
| Vulnerability Detected | VSFTPD Backdoor                                          |
| CVSS Score             | 10.0                                                     |
| Description            | Critical backdoor vulnerability detected in FTP service. |
| Risk Impact            | Remote attacker may gain unauthorized access.            |
| MITRE ATT&CK Technique | T1190 – Exploit Public-Facing Application                |
| Action Taken           | Vulnerability documented for remediation                 |
| Status                 | Open                                                     |
| Analyst Notes          | Immediate patching and service hardening recommended.    |

#### 6.3.4 APPENDIX D — OSQUERY PROCESS MONITORING REPORT

| Field                  | Value                                                   |
|------------------------|---------------------------------------------------------|
| Incident ID            | INC-004                                                 |
| Monitoring Tool        | Osquery                                                 |
| Event Type             | Suspicious Process Detection                            |
| Target System          | Windows VM                                              |
| Severity Level         | Low                                                     |
| Query Executed         | SELECT * FROM processes;                                |
| Description            | Newly created batch process detected during monitoring. |
| Detection Method       | Endpoint Process Monitoring                             |
| MITRE ATT&CK Technique | T1059 – Command and Scripting Interpreter               |
| Action Taken           | Verified process legitimacy                             |
| Status                 | Closed                                                  |
| Analyst Notes          | Simulated suspicious activity detected successfully.    |

#### 6.3.5 APPENDIX E — WIRESHARK TRAFFIC ANALYSIS REPORT

| Field           | Value                    |
|-----------------|--------------------------|
| Incident ID     | INC-005                  |
| Monitoring Tool | Wireshark                |
| Event Type      | Network Traffic Analysis |

| Field                  | Value                                                                      |
|------------------------|----------------------------------------------------------------------------|
| Protocols Observed     | TCP, UDP, ICMP, DNS, HTTP                                                  |
| Severity Level         | Informational                                                              |
| Description            | Network packet capture performed to analyse communication patterns.        |
| Indicators Observed    | DNS requests and ICMP ping traffic                                         |
| MITRE ATT&CK Technique | T1040 – Network Sniffing                                                   |
| Action Taken           | Packet traffic reviewed and filtered                                       |
| Status                 | Completed                                                                  |
| Analyst Notes          | Traffic analysis provided visibility into network communication behaviour. |

### 6.3.6 APPENDIX F — QRADAR EVENT ANALYSIS REPORT

| Field                  | Value                                                                             |
|------------------------|-----------------------------------------------------------------------------------|
| Incident ID            | INC-006                                                                           |
| Monitoring Tool        | IBM QRadar                                                                        |
| Event Type             | Security Event Investigation                                                      |
| Severity Level         | Medium                                                                            |
| Source IP Address      | 192.168.1.20                                                                      |
| Event Observed         | Failed Login Attempts                                                             |
| Description            | QRadar identified suspicious repeated login activity from a monitored IP address. |
| Detection Method       | Log Activity Monitoring                                                           |
| MITRE ATT&CK Technique | T1110 – Brute Force                                                               |
| Action Taken           | Logs analysed and suspicious activity reviewed                                    |
| Status                 | Investigated                                                                      |
| Analyst Notes          | SIEM filtering improved event investigation efficiency.                           |

### 6.3.7 INCIDENT RESPONSE SUMMARY TABLE

| Incident ID | Tool Used    | Severity | Status       |
|-------------|--------------|----------|--------------|
| INC-001     | Elastic SIEM | Medium   | Investigated |
| INC-002     | Snort IDS    | Medium   | Closed       |

| Incident ID | Tool Used         | Severity      | Status       |
|-------------|-------------------|---------------|--------------|
| INC-003     | Nessus Essentials | Critical      | Open         |
| INC-004     | Osquery           | Low           | Closed       |
| INC-005     | Wireshark         | Informational | Completed    |
| INC-006     | IBM QRadar        | Medium        | Investigated |

## 7. CONCLUSION

This project successfully demonstrated the implementation of a mini–Security Operations Center (SOC) lab environment for monitoring, detecting, and analysing cybersecurity events. Multiple industry-standard security tools including Wireshark, Snort IDS, Elastic SIEM, Kibana, IBM QRadar, Nessus Essentials, and Osquery were configured and tested in a controlled virtual environment.

The project provided practical exposure to packet analysis, intrusion detection, vulnerability assessment, log management, endpoint monitoring, and incident response workflows. Security events such as failed login attempts, suspicious network traffic, and simulated attack activities were successfully detected and investigated using SIEM dashboards and custom alert rules.

The implementation also highlighted the importance of centralized logging, continuous monitoring, and proactive threat detection in modern cybersecurity operations. MITRE ATT&CK mapping and incident response procedures further improved the understanding of real-world SOC analyst responsibilities.

Overall, this project enhanced practical cybersecurity knowledge and demonstrated how different security tools work together to improve organizational security posture and incident handling capabilities.

## 8. FUTURE ENHANCEMENTS

The following improvements can be implemented in future versions of the project:

- Integration with cloud-based SIEM platforms
- AI-based anomaly detection
- SOAR automation for incident response
- Real-time email alert notifications
- Ransomware behaviour detection
- Threat intelligence feed integration
- Active Directory monitoring
- Machine learning for suspicious activity prediction



- Advanced threat hunting dashboards
- Automated malware analysis

These enhancements can improve automation, detection accuracy, and SOC operational efficiency.

## 9. SUMMARY

The primary objective of this project was to design and implement a mini–Security Operations Center (SOC) lab environment capable of monitoring, detecting, and analyzing cybersecurity events in real time. The project focused on understanding SOC operations, security monitoring techniques, incident response procedures, log management, intrusion detection, and vulnerability assessment using industry-standard security tools.

The project was divided into theoretical and practical components. The theoretical section covered SOC fundamentals, SIEM technology, security monitoring concepts, log management processes, incident response lifecycle, documentation standards, and core cybersecurity principles such as the CIA Triad, defense-in-depth, and zero trust architecture.

The practical implementation involved configuring and testing multiple security tools including Wireshark, Snort IDS, Nessus Essentials, Elastic SIEM, Kibana, IBM QRadar, and Osquery within a controlled virtual lab environment. Network traffic analysis was performed using Wireshark to capture and analyze protocols such as TCP, UDP, ICMP, DNS, and HTTP. Intrusion detection was implemented using Snort with custom IDS rules to identify suspicious HTTP requests.

Vulnerability scanning was performed using Nessus Essentials against the Metasploitable2 virtual machine to identify critical security weaknesses and outdated services. Log collection and normalization activities were conducted using Fluentd and Logstash, while Elastic SIEM and Kibana were used to visualize and analyze security events. Osquery was used for endpoint monitoring and process visibility on a Windows virtual machine.

Additionally, simulated security incidents such as failed login attempts and suspicious network traffic were analyzed to understand SOC workflows including detection, triage, investigation, containment, eradication, and recovery. The project also incorporated MITRE ATT&CK mapping to relate observed activities with real-world adversary techniques used in cybersecurity attacks.

Overall, this project provided practical exposure to SOC operations, SIEM platforms, intrusion detection systems, log analysis, vulnerability assessment, and incident response procedures. The implementation demonstrated how multiple cybersecurity tools work together to improve organizational security monitoring, threat detection, and incident handling capabilities in modern enterprise environments.

## 10. REFERENCES

- [NIST SP 800-61 Revision 2 – Computer Security Incident Handling Guide](#)
- [MITRE ATT&CK Framework](#)
- [Snort IDS Documentation](#)
- [Elastic SIEM Documentation](#)
- [IBM QRadar Documentation](#)
- [Nessus Essentials Documentation](#)
- [Wireshark User Guide](#)
- [Osquery Documentation](#)
- [SANS Incident Handler's Handbook](#)